

MYSQL Constraints: -

Constraints can be specified when a table is created. Constraints are different types: -

- NOT NULL
- UNIQUE KEY
- PRIMARY KEY
- FOREIGN KEY
- COMPOSITE KEY
- CHECK
- DEFAULT
- AUTO_INCREMENT

1. **NOT NULL**: - The NOT NULL constraint enforces a field to always contain a value. This means that you cannot insert a new record, or update a record without adding a value to this field.
2. **UNIQUE KEY**: - The UNIQUE constraint uniquely identifies each record in a database table. The UNIQUE and PRIMARY KEY constraints both provide a guarantee for uniqueness for a column or set of columns. A PRIMARY KEY constraint automatically has a UNIQUE constraint defined on it. Note that you can have many UNIQUE constraints per table, but only one PRIMARY KEY constraint per table. UNIQUE key accept null values but PRIMARY KEY not accept null values.
3. **PRIMARY KEY**: - The PRIMARY KEY constraint uniquely identifies each record in a database table. Primary keys must contain unique values. A primary key column cannot contain NULL values. Each table should have a primary key, and each table can have only ONE primary key. The purpose is to bind data together, across tables, without repeating all of the data in every table.
4. **FOREIGN KEY**: - A FOREIGN KEY in one table points to a PRIMARY KEY in another table. The FOREIGN KEY constraint is used to prevent actions that would destroy links between tables. The FOREIGN KEY constraint also prevents that invalid data from being inserted into the foreign key column, because it has to be one of the values contained in the table it points to.
5. **COMPOSITE KEY**: - A composite key in MySQL is a combination of two or more than two columns in a table that allows us to identify each row of the table uniquely. It is a type of candidate key which is formed by more than one column. MySQL guaranteed the uniqueness of the column only when they are combined. If they have taken individually, the uniqueness cannot maintain.

Any key such as primary key, super key, or candidate key can be called composite key when they have combined with more than one attribute. A composite key is useful when the table needs to identify each record with more than one attribute uniquely. A column used in the composite key can have different data types. Thus, it is not required to be the same data type for the columns to make a composite key in MySQL.

6. **CHECK**: - The CHECK constraint is used to limit the value range that can be placed in a column.
7. **DEFAULT**: - The DEFAULT constraint is used to insert a default value into a column. The default value will be added to all new records, if no other value is specified.
8. **AUTO_INCREMENT**: - Auto-increment allows a unique number to be generated when a new record is inserted into a table.

MYSQL JOINS: -MySQL joins are used to query data from two or more tables, based on a relationship between certain columns in these tables.

Different MYSQL JOINS: -

- **CROSS JOIN:** That returns the Cartesian product of rows from the tables in the join. In other words, it combines each row from the first table with each row from the second table.
- **SELF JOIN:** Self join is a table joined to itself.
- **NATURAL JOIN:** Natural Join joins two tables based on same attribute name and datatypes.
- **INNER JOIN:** Return rows when there is at least one match in both tables.
- **LEFT JOIN:** Return all rows from the left table, even if there are no matches in the right table.
- **RIGHT JOIN:** Return all rows from the right table, even if there are no matches in the left table.
- **FULL JOIN:** Return rows when there is a match in one of the tables.

MYSQL DROP CONSTRAINT: -

(1.) DROP a PRIMARY KEY Constraint:-

```
>>ALTER TABLE TABLE_NAME DROP PRIMARY KEY;  
>> alter table emp drop primary key;  
>>desc emp;
```

(2.) DROP a UNIQUE KEY Constraint:-

```
>>ALTER TABLE TABLE_NAME DROP INDEX unique_key_column_name;  
>> alter table emp drop index id;  
>>desc emp;
```

(3.) DROP a NOT NULL Constraint:-

```
>>ALTER TABLE TABLE_NAME MODIFY not_null_column_name datatype NULL;  
>> alter table emp modify name varchar(20) null;  
>>desc emp;
```

(4.) DROP a CHECK Constraint:-

```
>>ALTER TABLE TABLE_NAME DROP CONSTRAINT constraint_name;  
>> alter table emp drop constraint constraint_1;  
>>desc emp;
```

(5.) DROP a DEFAULT Constraint:-

```
>>ALTER TABLE TABLE_NAME ALTER default_column_name DROP DEFAULT;  
>> alter table emp alter city drop default;  
>>desc emp;
```

(6.) DROP a AUTO_INCREMENT Constraint:-

```
>>ALTER TABLE TABLE_NAME DROP PRIMARY KEY, change  
auto_increment_column_name auto_increment_column_name datatype;  
>> alter table emp drop primary key,change id id int;  
>>desc emp;
```

(7.) DROP a COMPOSITE KEY Constraint:-

```
>>ALTER TABLE TABLE_NAME DROP PRIMARY KEY;  
>> alter table emp drop primary key;  
>>desc emp;
```

(8.) DROP a FOREIGN KEY Constraint:-

```
>>ALTER TABLE TABLE_NAME DROP FOREIGN KEY constraint_name;  
>> alter table orders DROP FOREIGN KEY orders_ibfk_1;  
>>desc emp;
```